

A LASSO Type Factor Model in Cryptocurrency

Jinchuan Li

Central University of Finance and Economics

Yifeng Zhu

Central University of Finance and Economics*

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Abstract: In this paper, we examine the performance of 49 cryptocurrency market anomalies in terms of their in-sample, out-of-sample, and full sample performance following the work of Liu et al. (2022). We find that, despite the similarities in anomaly performance between the in-sample and overall sample periods, there are noticeable changes in the behavior of out-of-sample anomalies. This includes the disappearance of size effect and the appearance of left-tail risk effect. Consequently, by using the Iterative Double Selection Lasso method, we construct a new three-factor model — DS3, which consists of the market factor (MKT), the two-week momentum factor (MOM2), and the residual momentum factor (RMOM). In comparison to CPT3 (Liu et al., 2022) and IPCA3 (Bianchi and Babiak, 2023), our DS3 model exhibits a certain advantage in explaining anomalies.

JEL Classification: G11, G12

Keywords: Cryptocurrency market, Anomalies, Iterative Double Selection Lasso

1. Introduction

Cryptocurrencies are initially developed to address real-world transaction issues, such as reducing costs and tracking transactions (Nakamoto, 2008). However, their innovative nature and remarkable volatility transform them into an appealing investment and financing market. Since 2014, the cryptocurrency market witnesses rapid growth and extreme fluctuations, characterized by vast trading volumes, numerous ICOs and de-listings, and a diverse array of traders. Consequently, the cryptocurrency market evolves into a speculator's haven, where the prices of various currencies are subject to fluctuations influenced by factors like cryptocurrency issuance, the expansion of exchanges, and a diverse trader base. Investigating the factors that impact expected returns across the entire cryptocurrency market is an urgent area of study for both scholars and traders.

Scholars have conducted research on cross-sectional asset pricing in the cryptocurrency market and find a lot of anomalies. Liu et al. (2022) find significant momentum and size effects in their study from January 2014 to July 2020 and find size-related anomalies and momentum related anomalies show effective predictive power. Zhang et al. (2021) study the relationship between left-tail risk and expected returns from 2016 to 2020 and found that left-tail risk can positively predict expected returns cross-sectionally for this period. Cai and Zhao (2024) open up behavioral finance research on cross-sectional pricing in the cryptocurrency market by studying the salience effect. Dong et al. (2022) test cross-sectional anomalies in the cryptocurrency market at a monthly frequency from 2014 to 2019 and find that a decrease in liquidity enhances the performance of anomalies. Jiang et al. (2023) find the macroeconomic factors such as consumption, uncertainty, and sentiment play an important role in explaining the deviation from fundamentals in cryptocurrency.

In the realm of equities, as the number of anomalies has steadily grown, scholars have started leveraging machine learning techniques to tackle these high-dimensional anomalies (the “factor zoo”, Harvey et al., 2016; Hou et al., 2020; Jensen et al., 2021) for more accurate predictions of asset returns (Gu et al., 2020a). Building upon this, some studies have used models with stronger generalization capabilities to capture the more intricate information within high-dimensional anomalies (Cong et al., 2020; Gu et al., 2020a), while others have resorted to “dimension reduction” methods or added penalty terms to linear models to extract the core factors that drive asset prices (Gu et al., 2020a; Kelly et al., 2019; Gu et al., 2020b; Freyberger et al., 2020). Beyond prediction, another worthwhile application of machine learning methods is to more scientifically pick factors with potential explanatory power for returns from the vast array of pricing anomalies. Since factors often share economic and statistical correlations, research indicates that in high-dimensional settings, testing individual

factors tends to produce biases (Belloni et al., 2012), and these biases can in turn affect the results of such factor tests. Meanwhile, the large model variance in high-dimensional scenarios makes it hard to carry out effective statistical inference. Feng et al. (2020) put forward that under the linear SDF framework, the Double-Selection Lasso can achieve asymptotically unbiased estimation of incremental factors within a high-dimensional set of factors. Using machine learning methods to select effective pricing factors not only avoids the subjectivity of manual selection but also allows the chosen factors to better explain anomalies in financial markets, thereby forming an effective factor model.

In the realm of cryptocurrency, the exploration of factor models remains limited. Liu et al. (2022) devise factor combinations — CMKT, CSMB, and CMOM — providing rationale through a PCA (Principal Component Analysis) approach. Bianchi and Babiak (2023) suggest utilizing the IPCA (Instrumented Principal Component Analysis) method to formulate a latent factor model for the cryptocurrency market. Nevertheless, there is currently no universally accepted econometric methodology for constructing factor models that are specifically designed for the nuances of the cryptocurrency market. Feng et al. (2020) propose the Double-Selection Lasso to assess the contribution of new factors to existing factor models or factor libraries. The main issue addressed is how to select suitable factors from an existing factor library as explanatory variables to verify whether new factors have new sources of returns. Inspired by this, Chen et al. (2023) apply Iterative Double-Selection Lasso to construct a Global Factor Model, addressing stock asset pricing issues on a global scale.

In this study, we re-examine 49 anomalies in the cryptocurrency market and find that only 13 of them are significant from 2014 to 2023, which is consistent with Liu et al. (2022). However, the left-tail risk and salience group anomalies, which was respectively founded by Zhang et al. (2021) and Cai and Zhao (2024), are no longer significant during the full sample period. Additionally, our study employs Iterative Double-Selection Lasso to establish a factor model based on the anomalies constructed earlier. The DS3 model (which consists of the market factor, the two-week momentum factor, and the residual momentum factor) constructed in this study is compared with the cryptocurrency three-factor model proposed by Liu et al. (2022). The results indicate that, within the sample, our constructed DS3 model performs as well as Liu et al.'s (2022) factor model (referred to as CPT3 henceforth). However, out-of-sample, the model we constructed demonstrates even more superior performance. Furthermore, we investigated the factors contributing to the subpar performance of CPT3 over an extended period. Our study unveiled during out-of-sample period, CPT3 factors were ineffective in explaining left-tail risks and the size factor is not significant anymore. Integrating these

insights with the outcomes of the Double-Selection method, we proposed the DS3 model, which, at its core, possesses the capability to comprehensively address all anomalies in the market. Notably, our factor model comparison also incorporates the IPCA3 model (a three-factor model built via IPCA), though IPCA is not inherently designed for explaining anomalies. The results indicate that while IPCA3 reduces anomaly magnitudes, it remains less effective than DS3 in explaining and eliminating them. At Last, we conduct robustness tests, which includes encompassing stability assessments of different anomalies through WLS-FM regression, the creation of anomalies based on cryptocurrencies with large market capitalization to evaluate factor model explanatory power, and adjustments to the factor construction method of CPT3 to alleviate the impact of grouping on factor explanatory power. These tests collectively demonstrate the robustness of our findings.

Our work makes two primary contributions to the literature on cryptocurrency market anomalies. First, we extend the work of Liu et al. (2022) by re-examining common anomalies in cryptocurrency markets using updated data, documenting significant changes in anomaly patterns - notably, the previously documented size effect appears to have diminished or disappeared entirely in more recent periods. Second, and more importantly, we develop a novel DS3 factor model constructed using Liu et al.'s original sample period data; crucially, this model demonstrates remarkable out-of-sample robustness as it continues to perform well when applied to post-Liu et al. periods, while also exhibiting superior anomaly-elimination capabilities compared to existing models. These findings suggest that our DS3 model represents a significant advancement in modeling cryptocurrency returns, offering both theoretical and practical improvements over current approaches.

The rest of this paper is structured as follows: In Section 2, we illustrate the data processing steps and the anomalies we construct as well as the principle of Double Selection Lasso (DS Lasso). In Section 3, we initially conduct single sorting tests based on different detection periods. Subsequently, we employ the DS-Lasso method to construct the DS3 factor model and compare its performance with CPT3. The fourth section is dedicated to discussing the underperformance of CPT3. In Section 5, significance tests were carried out. Section 6 provides a summary of the entire text.

2. Data and Methodology

2.1 Data and data preprocessing

Our data includes daily closing prices, which we use to calculate daily returns, market capitalization, and

dollar trading volume. Following Liu et al. (2022), our study relies on trading data sourced from CoinMarketCap (<https://interest.coinmarketcap.com/>), which is a well-respected website for cryptocurrency information. Since cryptocurrencies lack a centralized trading venue, CoinMarketCap solves this issue by calculating a value-weighted price for each currency across various exchanges, providing a comprehensive price for each cryptocurrency.

As noted by CoinMarketCap, specific market pairs are excluded when calculating cryptocurrency prices to ensure the reliability of the data. These exclusions include: (1) cases where prices do not reflect free market conditions—for example, when an exchange disables withdrawals or deposits, or regulatory constraints restrict purchases to a specific geographical region, limiting broader market participation; (2) instances where its algorithms identify reported prices as significant outliers relative to other market pairs for the same cryptocurrency; and (3) market pairs on exchanges that either do not enforce trading fees or offer substantial trading incentives. Such market pairs are particularly susceptible to wash trading, which can artificially inflate reported volumes. CoinMarketCap explains that excluding these markets is intended to provide a more accurate representation of relative trading volumes across the cryptocurrency market. Meanwhile, to avoid survivorship bias, we iterated through all `coin_ids` when retrieving the data — these `coin_ids` are used by CoinMarketCap to uniquely identify cryptocurrencies.¹

While CoinMarketCap provides accurate and comprehensive trading data, it still contains some NA values. Therefore, we exclude observations with zero market or transaction value, as well as any missing values for market capitalization, trading volume, or closing price. We also remove one currency that briefly exceeded Bitcoin’s market capitalization in October 2021 due to abnormal returns. After cleaning the raw data, we calculate the daily return for each currency and subsequently the weekly return based on daily returns.

To divide trading days into weeks, we adopt the approach used by Liu et al. (2022). We designate the first natural day of each year as the first day of the first week and generate 51 seven-day weeks. The last week of each year consists of 8 or 9 days, depending on the number of remaining trading days. We exclude currencies with a market capitalization less than \$1 million by the end of each week, ensuring they are not included in our analysis. Our datasets comprises 3,969 currencies and covers the period from January 1, 2014, to December 31, 2023.

Table 1 summarizes our sample, which shows that the number of cryptocurrencies that meet all the filters has increased from 121 in 2014 to 2,622 in 2021 and then decreased to 1,827 in 2023. Over the ten-year period,

¹ See <https://support.coinmarketcap.com/hc/en-us/articles/360043395752-Price-Market-Pair-Cryptoasset>

the cryptocurrency market has gradually expanded. The average market value of currencies grows from \$232.58 million in 2014 to \$924.20 million in 2023, while the average daily dollar volume increases from \$1.230 million in 2014 to \$47.626 million in 2023. Notably, the growth in dollar volume exceeds the growth in market capitalization, indicating the highly volatile nature of the crypto market. We also observe a left-skewed distribution of market capitalization and trading volume, with significantly lower median values than the average values, suggesting the presence of outliers in the market.

[Insert Table 1 about here]

2.2 Constructing anomalies

Due to the lack of financial data in the cryptocurrency market, cross-sectional characteristics can only be constructed based on trading data. Previous studies by Liu et al. (2022) have identified significant momentum and size effects in the cryptocurrency market. Additionally, trading volume and the standard deviation of trading volume have been found to be significant anomalies. Zhang et al. (2021) analyze left-tail risk using data from the past 90 days and found a positive relationship between left-tail risk and returns within their specified sample interval. Cai and Zhao (2024) investigate the salience effect in cryptocurrencies and observed a negative relationship between the salience price and expected returns, providing insights from the perspective of behavioral finance.

In this study, we construct 49 anomalies in the cryptocurrency market by drawing upon cross-sectional anomalies observed in the stock market. These anomalies are classified into seven categories, Size, Momentum (Mom), Trading Volume (Volume), Volatility (Vol), Downside Risk, Left-tail Risk, and Salience. The sources and methods used to construct these anomalies are displayed in Table 2.

[Insert Table 2 about here]

2.3 Selection model: Iterative double selection lasso

According to Ross's (1976) APT model, the expected return of an asset can be expressed in the following form, where $E(r_i^e)$ represents the expected excess return of asset i , f_k represents the expected return of factor k , and β_{ik} represents the exposure of asset i to factor k .

$$E(r_i^e) = u_i - r_f = \beta_{i1}f_1 + \beta_{i2}f_2 + \cdots + \beta_{ik}f_i$$

If we want to assess the suitability of new factors within an existing factor model, the newly constructed factor model is:

$$E(r_i^e) = \alpha_i + \beta_h h_t + \beta_g g_t$$

Specifically, according to the derivation by Feng et al. (2020), the above content can be transformed into the following form:

$$E(r_i^e) = \ln\gamma_0 + C_h\lambda_h + C_g\lambda_g$$

Where C_h and C_g represent the covariances between factors and their expected returns, λ_h and λ_g represents the excess risk premium associated with the factors. In this context, factors enhance the explanatory capability of the existing factor set, akin to having substantial risk premiums. Given the potential size of the existing factor set, direct regression methods may result in high-dimensional econometric challenges. Consequently, Feng et al. (2020) ingeniously employ the Double Selection research framework to identify factors potentially correlated with those from the existing factor set h_t . These chosen factors are then integrated into the model, enabling the determination of the genuine excess risk premium of the testing factor g . The Double-Selection Lasso utilized in this study is an adaptation by Chen et al. (2023) aimed at addressing certain issues in the two-stage selection process. The specific steps are outlined as follows:

(Step 1a) Calculate the average return of assets in the sample, denoted as $\bar{r}_t$, using existing factors as explanatory variables and the average return as the dependent variable. Conduct the following Lasso regression to obtain factors with non-zero coefficients, collectively denoted as I_1 . This step is undertaken to select factors with potential correlations to expected returns.

$$\bar{r}_t = \ln\gamma_0 + C_h\lambda_h$$

(Step 1b) Utilize existing factors as explanatory variables and the factor under examination as the dependent variable. Conduct Lasso regression to obtain factors with non-zero coefficients, denoted as I_2 . Essentially, this step involves selecting factors from the existing set that are correlated with the factor under examination.

$$g_t = \alpha + \beta_h h_t + \epsilon$$

(Step 2) Compute the union of sets I_1 and I_2 , denoted as I_3 . Employ the factors from I_3 as control variables, with g as explanatory variables and as the $\bar{r}_t$ dependent variable. Conduct an OLS regression to obtain coefficients of g , representing the risk premium, and simultaneously calculate the t-statistics for these coefficients.

$$\bar{r}_t = \ln\gamma_0 + C_{\{I_3\}}\lambda_{\{I_3\}} + C_g\lambda_g$$

Repeat the above process, with the initial set as the market factor. For each iteration, begin with a round of (Step 1a), followed by sequentially selecting anomalies from the 49 anomalies constructed in this study for (Step 1b) and (Step 2). Collect the risk premiums and t-statistics corresponding to each anomaly, select the

anomaly with the maximum t-statistic, and incorporate it into the formation of a new factor model. Continue this cycle until the risk premiums of the detected anomalies are not statistically significant. Due to the likelihood of having numerous factors in the final result, we adopt a stepwise regression method to further reduce the number of factors and obtain a new factor model. Initially, we eliminate variables with factor return t-statistics less than 1.64. Next, we conduct stepwise regression, choosing CAPM as the initial factor model. In each iteration, sequentially select factors from as the dependent variable, use the current model to explain them, obtain the risk premiums and t-statistics for each anomaly, and incorporate the anomaly with the maximum t-statistic into the factor model.

3. Empirical Results

3.1 Single sorting test

Following Liu et al. (2022), we employ a single sorting test to construct portfolios based on 49 cross-sectional features in the cryptocurrency market. These features are categorized into seven categories, namely size, momentum, trading volume, volatility, downside risk, left-tail risk, and salience. At the end of each week, we sort the cryptocurrencies into five groups based on the quantile of each characteristic and calculate the value-weighted returns for each portfolio. To construct the long-short portfolio, we take the difference between the highest and lowest groups. The Newey-West method is used to obtain adjusted t -values of the long-short portfolio return series, and we observe whether these return series are significantly different from zero. To enhance the examination of anomalies in the cryptocurrency market across diverse time frames, we independently assessed the performance of 49 anomalies during three distinct periods: January 2014 to July 2020, August 2020 to December 2023, and January 2014 to December 2023. The initial phase aligns with the period when Liu et al. (2022) identified anomalies and is denoted as the “in-sample period” in relation to our study. The subsequent phase is labeled the “out-of-sample period,” and the third phase is termed the “full-sample period.”

Table 3 presents the mean returns and t-test outcomes for both long and short combinations, formed utilizing 49 features across three distinct time frames. Notably, the peculiar performance of short-term momentum showcases remarkable consistency. Throughout the in-sample phase, one-week momentum, two-week momentum, three-week momentum, four-week momentum, and one-to-three-week momentum all

exhibit significance at the 5% level, yielding respective average weekly returns of 3.6% ($t=2.635$), 4.0% ($t=3.779$), 4.0% ($t=3.831$), 3.1% ($t=3.065$), and 2.6% ($t=2.440$). This implies that currencies demonstrating robust recent performance maintain positive trends in the future. Moreover, it is worth highlighting that this phenomenon persists in the remaining two periods. For example, in the out-of-sample phase, two-week momentum continues to generate a notable weekly return of 3.8% ($t=4.943$), and over the entire period, two-week momentum sustains a return of 3.9% ($t=5.322$) per week. Surprisingly, residual momentum also exhibits substantial significance and consistency. In the in-sample period, the residual momentum anomaly yields an average weekly return of 2.1%, with a t-value of 2.422, signifying significance at the 5% level. Although it diminishes in the out-of-sample period, with returns decreasing to 1.2% per week and a corresponding t-value of 2.378, it remains statistically significant. Overall, across the entire sample, residual momentum averages a return of 1.8% ($t=3.046$), also significant at the 5% level. Regrettably, despite the significance and consistency of residual momentum across various sample periods, this analysis does not reveal a distinct superiority of residual momentum over traditional momentum. Similar conclusions have been evidenced in the context of the stock market, as Blitz et al. (2011) found that in the stock market, residual momentum demonstrates superior stability and risk-adjusted returns compared to traditional momentum.

Analyzing the characteristics of anomaly returns within and outside the sample, we uncover intriguing insights. Initially, anomalies associated with size exhibit significance in the in-sample phase. Notably, MACP, representing the logarithm of cryptocurrency market capitalization, demonstrates noteworthy returns in the constructed long-short portfolio, averaging -2.5% per week with a t-value of -3.711, signifying significance at the 1% level. Similarly, related anomalies such as PRC (price at the end of the last week), MAXPRC (maximum price during the last week), PRCVOL (based on trading volume in dollars), and STDVOLSCALED (standard deviation of trading volume in dollars) also manifest substantial returns during the in-sample period. The returns and t-values for portfolios built upon these anomalies are -2.4% ($t=-2.088$), -2.5% ($t=-2.073$), -2.4% ($t=-2.495$), and -2.7% ($t=-3.025$), respectively, all statistically significant at the 5% level. The consistent trend of these anomalies implies that smaller market cap currencies are likely undervalued or predisposed to yield higher returns in the subsequent week compared to their larger counterparts. However, intriguingly, during the out-of-sample period, these size-related anomalies lose their significance. For example, MACP only achieves a return of -0.1%, with a reduced t-value of -0.296, failing to reject the null hypothesis at the 5% significance level. Similarly, we observe returns and t-values of -0.9% ($t=-1.116$) for PRC, -0.7% ($t=-0.954$) for MAXPRC, 0.3% ($t=0.675$) for PRCVOL, and -0.3% ($t=-0.106$) for

STDVOLSCALED, none of which reach statistical significance at the 5% level. Nevertheless, overall, size-related anomalies persist across the entire sample, with MACP averaging a return of -1.7% and a t-value of -3.617, signifying significance at the 5% level. The performance of other anomalies is comparable, and over an extended time period, the overall sample results align with the findings of Liu et al. (2022) during the backtesting interval (i.e., the in-sample period).

Additionally, during the out-of-sample period, anomalies measuring the absolute level of historical left-tail risk in the out-of-sample left-tail risk category and the SVAR (semi-variance) in downside risk become significant. Among these anomalies characterizing left-tail risk and left-skewed features, currencies with higher risk tend to be relatively overestimated, resulting in relatively smaller relative returns in the following week. According to our results, the maximum spread is observed in the MIN anomaly, with a performance of -2.5% ($t=-3.990$), and the spread is seen in the DES10 anomaly, with a performance of 0.1% ($t=0.193$). During this period, the performance of anomalies in the cryptocurrency market contradicts the findings of Zhang et al. (2021). They examined the cryptocurrency market from January 2016 to December 2020, calculated a 90-day value at risk (VaR), and found that VaR had a positive predictive ability for expected returns. This discrepancy may be due to differences in our calculation methods and calculation intervals. In the next section, we will test the cross-week predictive ability of anomalies, and the results of this test are consistent with Zhang et al. (2021). In the in-sample and full-sample periods, extreme risk and left-tail risk did not show significant predictive ability for expected returns. However, during the out-sample period, DVAR1 and DES1 could each achieve average weekly returns of 1.3% and 1.3%, respectively, and were statistically significant at the 5% level.

To summarize, it can be deduced that the pricing logic in the cryptocurrency market may have undergone substantial changes during the out-of-sample period. While the primary significant anomalies do not display inherent distinctions, the efficacy of the original factor model in pricing and elucidating anomalies might diminish after this timeframe. In light of this, we suggest conducting additional research to update the factor model.

[Insert Table 3 about here]

3.2 Long-term exam

In this subsection, we aim to test the long-term cross-sectional predictive ability of various characteristics on expected returns by constructing investment portfolios and calculating their returns over the future six weeks. Subsequently, we perform statistical inference on the time series of these returns. Our analysis aims to

identify any anomalies that can predict cryptocurrency returns cross-sectionally and investor behavior over extended periods. If the previously significant characteristics remain significant in the test, it indicates that these anomalies can predict cryptocurrency returns cross-sectionally and investor behavior in the long term. If previously insignificant characteristics become significant in the test, we believe that these features can predict across periods. However, some mechanisms in the market may delay the predictive ability of these characteristics, possibly due to insufficient information transmission in the market.

Table 4 presents the results of our long-term predictive ability tests. Our findings indicate that market capitalization, price, highest close price of week, dollar trading volume, the standard deviation of trading volume, and the Amihud ratio have relatively stable predictive ability for returns cross-sectionally, which may partly result from the size effect. In the cryptocurrency market, the difference in trading volume between different currencies is significant due to the large range of currency market capitalization. Therefore, the significant returns of the volume or *STDPRCVOL* long-short portfolio are attributable to the size effect. Moreover, when calculating the Amihud ratio, dollar trading volume appears as the denominator, which may explain why Amihud can earn positive returns.

[Insert Table 4 about here]

Furthermore, we find that the conventional momentum and residual momentum have poor long-term cross-sectional predictive power for expected returns. Only the one-week return is a significant anomaly at the $t+3$ period. As time increases, the cumulative return gradually becomes a negative cross-sectional predictor of expected returns, and some cumulative returns can significantly negatively predict expected returns.

We also observe that the left-tail risk, has cross-period predictive ability for expected returns. Although they are not a significant predictor of expected returns at the $t+1$ period, it can predict expected returns significantly cross-period, which is consistent with the conclusion of Zhang et al. (2021). We calculate the ES5 transfer matrix to assess the differences between cryptocurrencies in the 1 to 5 investment portfolios at the end of the t period and $t+k$ period. Our results indicate that ES5 exhibits little change with a one-period interval, while there is a significant change with a three-period interval, indicating that investor reactions have lagged. This delay is also the reason why ES5 does not have cross-sectional predictive ability for expected returns at the $t+1$ period. Additionally, our findings indicate that both *VOLSCALED* and *STDVOLSCALED* exhibit lagged predictive power across periods, implying the presence of a potential delay in the information transmission mechanism within the cryptocurrency market

3.3 Construct factor model using DS lasso

In this section, we employ the Double-Selection Lasso methodology to construct a novel factor model based on the examined characteristics. Following the approach of Chen et al. (2023), we compute factor returns using a three-group sorting method. Specifically, we sort stocks into 30%-40%-30% groupings and calculate the value-weighted return spread between the top and bottom groups. To compare the factor model selected by Double Selection Lasso with the three-factor model constructed by Liu et al. (2022), we use data from January 2014 to July 2020 for factor selection with Double Selection Lasso and employ data from August 2020 to December 2023 as out-of-sample data to compare the out-of-sample explanatory power of different factor models. The final factor model (denoted as DS3 model) selected consists of MKT, MOM2, and RMOM. MKT represents the market value-weighted returns of all cryptocurrencies in the sample; MOM2 denotes the two-week momentum factor; RMOM is the residual momentum factor, calculated as the four-week cumulative residuals of regressing coin returns on the CAPM over the prior 4 weeks from week $t-4$ to week t . To eliminate the correlation between factors, we orthogonalize the factors following the approach of Liu et al. (2022)². Specifically, when constructing the RMOM factor, we first classify cryptocurrencies into winners and losers based on their returns over the past three weeks. Subsequently, within each group, we further divide the currencies into three parts based on the values of the RMOM feature—30%, 40%, and 30%, respectively. Finally, the residual momentum factor's returns are calculated using the following formula:

$$RMOM = \frac{1}{2}(winner\ high + loser\ high) - \frac{1}{2}(winner\ low + loser\ low)$$

Table 5 displays the descriptive statistics of the DS3 model before orthogonalization, the DS3 model after orthogonalization, and Liu et al.'s (2022) CPT3 model. From this, we observe that prior to orthogonalization, MOM2 and Raw RMOM exhibit a relatively high correlation of 0.31 due to being in the same category. Post-orthogonalization, although the average returns and t-values of RMOM decrease compared to pre-orthogonalization, the correlation between RMOM and MOM2 significantly decreases to only 0.15. In the following sections, we will use MKT, MOM2, and the orthogonalized residual momentum factor (RMOM) to construct the DS3 factor model and compare it with the CPT3 model.

[Insert Table 5 about here]

3.4 Test factor models

In this section, we compare the CAPM model, CPT3 model, DS3 model, and IPCA3 model across

² Unlike the orthogonalization procedure in Fama and French (1993), when constructing the CPT3 factor model, Liu et al. (2022) only orthogonalized the CMOM factor but did not orthogonalize the CSMB factor. Here we follow Liu's approach by orthogonalizing only RMOM and not orthogonalizing MOM2.

different dimensions. It should be noted in advance that the factors of the CPT3 and DS3 models were selected using in-sample period data, whereas the factor returns of the IPCA model are derived from model estimation, meaning the IPCA3 model is estimated using full-sample period data. Through comparison, we find that the DS3 model outperforms the CPT3 model in both the out-of-sample and full-sample periods. This indicates that when constructing factor models from anomalies, the Double Lasso-selected model exhibits better out-of-sample performance than the intuitively selected factor model. Moreover, compared to the IPCA3 model, the DS3 model demonstrates superior anomaly-elimination capability, further highlighting its advantages.

3.4.1 In-sample test

From 3.4.1 to 3.4.3, we will compare the explanatory power of different factor models. Following Liu et al. (2019), we use the 49 anomalies as the dependent variable and different combinations of factors as the independent variables. We observe the average absolute values, average absolute T-statistics, the p-value of the GRS statistic in the regression equations, the number of minimum alphas which is the number of anomalies for which the factor model produces the smallest absolute alpha among the models being compared in the table and the number of unexplained alphas which is the number of anomalies with t-values of the factor models' alphas greater than 1.96.

Table 6 presents a comparative analysis of the four factor models. In the specific context of the in-sample period, as elucidated by the Capital Asset Pricing Model (CAPM), the mean absolute alpha for anomalies stands at 2.727% (annualized to 141.804%), accompanied by a T-statistics mean absolute value of 2.785. Conversely, subsequent to the application of the CPT3 model for explanation, anomalies lose their overall significance, revealing a p-value of 0.131 for the Generalized R-squared (GRS) test, where the null hypothesis posits anomalies jointly equal to 0. This failure to reject the null hypothesis at a 5% significance level indicates that, within the sample period, the CPT3 model adeptly elucidates existing anomalies. Furthermore, it is notable that CPT3 diminishes the absolute alpha value for anomalies to an average of 0.972% (annualized at 50.544%), with the average T-statistics value reaching only 1.291. Interestingly, as for the IPCA3 model, among the four models, IPCA3 has the smallest mean absolute alpha, reaching 0.120, but its average t-value is as high as 3.726. This indicates that although the model can minimize the scale of abnormal returns to some extent, the IPCA model cannot systematically eliminate anomalies during in-sample period.

During the in-sample period, the explanatory capability of the DS3 model is inferior to that of the CPT3 model, although still effective. The GRS test p-value for anomaly returns following DS3 explanation is 0.126,

smaller than CPT3's test results; however, at the 5% level, it lacks statistical significance. Essentially, there is no reason to reject the null hypothesis that anomaly returns, post-DS3 adjustment, jointly equal zero. Moreover, subsequent to the DS3 test, the mean absolute value of anomaly returns is 1.011% (annualized at 52.573%), and the average t-statistic is 1.317, slightly larger than CPT3's statistics but not significant at a 5% level.

3.4.2 Out-of-sample test

We employed data spanning from August 2020 to December 2023 as an out-of-sample validation set to evaluate the predictive performance of various factor models. To begin, akin to the results observed within the sample, the Capital Asset Pricing Model (CAPM) displayed lackluster performance during the out-of-sample period, primarily attributable to the constrained information within the factor model. The adjusted absolute average return for anomalies stood at 1.950% (annualized at 101.400%), accompanied by an absolute average T-statistic value of 3.056 and a GRS of 0.000. It can be asserted that the CAPM model struggles to effectively elucidate anomalies in the cryptocurrency market during the out-of-sample period. Moreover, the performance of the CPT3 model was suboptimal during this timeframe. Following CPT3 interpretation, the average alpha of anomalies remained elevated at 1.146% (annualized at 59.592%), with the absolute average T-statistic value for anomaly alpha reaching 2.075. It is worth noting that, after adjustment by the CPT model, the p-value of the GRS test for alpha is 0.000, indicating that at a 5% significance level, the model cannot explain all anomalies during the out-of-sample period. The IPCA3 model performed poorly at this stage. Although it could explain all anomalies, it had the largest mean absolute alpha.

Conversely, the DS3 model demonstrated resilient performance during this out-of-sample period. Post-DS3 interpretation, the absolute average of anomaly alpha was 0.592% (annualized at 30.784%), accompanied by an absolute average T-statistic value of 1.148. Simultaneously, the p-value of the GRS statistic is 0.027, indicating that at a 1% significance level, DS3 can explain all anomalies within the out-of-sample period. Although slightly elevated compared to the in-sample scenario, the DS3 model showcased commendable out-of-sample capabilities, taking into account both the magnitude of the T-statistic average (with a 5% significance threshold around 1.96) and comparisons among CAPM, CPT3 and IPCA3.

[Insert Table 6 about here]

3.4.3 Full-sample period

While the DS3 model surpasses CAPM and CPT3 in out-of-sample performance, its underperformance compared to CPT3 in the in-sample analysis hinders us from conclusively determining the superiority or inferiority between DS3 and CPT3 models. In Panel 3 of Table 6, where two time periods are amalgamated,

the scrutiny of factor models across distinct market cycles for their explanatory prowess on anomalies robustly underscores DS3's advantage in elucidating anomalies.

Within the entire sample, the CAPM model remains relatively feeble, displaying an absolute average value of anomaly alpha at 2.155% (annualized at 112.06%), a notable T-statistic value of 3.312, and a diminished GRS statistic with a p-value of 0.000. The CPT3 model performs admirably, featuring an average alpha value of 0.814% (annualized at 42.328%), an absolute average T-statistic value of 1.560, below the 5% significance threshold of 1.96. However, the corresponding GRS test statistic's p-value is 0.008, less than 0.05, leading to the rejection of the null hypothesis at the 5% significance level. This indicates that CPT3 may struggle to explain all significant anomalies under more stringent conditions. Similar to the in-sample period, IPCA3 performs well in reducing the scale of anomalies but struggles to eliminate their significance, failing to explain 9 anomalies while maintaining a high average t-value of 3.529. Conversely, the DS3 model, post-explanation of anomalies, exhibits an absolute average value of anomaly alpha at 0.791% (annualized at 38.532%), with an absolute average T-statistic value of 1.338. Both values are comparatively diminutive among the four models. Furthermore, the GRS test p-value is 0.061, marginally exceeding 0.05. Under this more rigorous criterion, it can be inferred that the DS3 model possesses the capability to elucidate significant anomalies in the market. For robustness checks, the results of replicating the analyses in Table 6 with data sourced from CryptoCompare are presented in Table A1 of the Appendix.

3.4.4 Factor mutual explanations test

In this section, following the approach of Liu et al. (2022) comparing CH3 and FF3 factors, we investigate CPT3, DS3 and IPCA3 by mutually explaining their unique factors. Our findings indicate that the DS3 factors effectively explain the factors in both CPT3 and IPCA3. Under the DS3 model, the alpha values for the CSMB and CMOM factors are 71.2 basis points (annualized at 37.02%) and 7.5 basis points (annualized at 3.90%), with t-values of 1.166 and 0.207, respectively. Neither is statistically significant at the 5% level (critical t-value ≈ 1.96). Meanwhile, the adjusted alphas for the three IPCA factors are 4.697% ($t = 0.103$), 1.742 ($t = 0.688$), and 3.564 ($t = 1.989$). Except for the last factor, the DS3 model successfully explains the other two IPCA3 factors. Additionally, in the GRS test for the joint alpha of CSMB and CMOM being zero, the p-value is 0.897, indicating that, at the 5% significance level, we cannot reject the null hypothesis. The same applies to the IPCA3 model. Under the explanation of the DS3 model, the three factors of the IPCA3 model yield a GRS statistic of 1.841 with a p-value of 0.139. Therefore, we conclude that the DS3 model can effectively

explain the information contained in CPT3 and IPCA3.

In contrast, CPT3 fails to effectively explain the factors in the DS3 model. After controlling for the CPT3 model, RMOM retains an alpha of 89.4 basis points (annualized at 46.88%) with a t-statistic of 2.579, which is statistically significant at the 5% level. Similarly, MOM2 maintains an alpha of 147.6 basis points (annualized at 76.752%) with a t-statistic of 3.482, also significant at the 5% level. The GRS test yields a p-value of 0.000, rejecting the null hypothesis at the 5% significance level that the joint adjusted returns of RMOM and MOM2 are zero. Collectively, these results suggest that DS3 incorporates information not captured by CPT3.

Similarly, IPCA3 also fails to fully account for the DS3 factors. After adjusting for the IPCA3 model, RMOM still exhibits an alpha of 95 basis points with a t-statistic of 2.471, significant at the 5% level, while MOM2 shows an alpha of 255 basis points with a highly significant t-statistic of 2.553 (5% level). The GRS test p-value of 0.001 leads to rejection of the null hypothesis at the 5% level, confirming that DS3 contains pricing information beyond what IPCA3 can explain.

In summary, both CPT3 and IPCA3 are unable to fully explain the factor structure embedded in the DS3 model, highlighting the incremental explanatory power of DS3.

[Insert Table 7 about here]

4. Further Analysis:

4.1 Why is there a decline in the explanatory power of the CPT3 model?

In the preceding section, we established that the CPT3 model adeptly elucidates anomalies within the in-sample data and demonstrates satisfactory performance. Nevertheless, in the out-of-sample phase, its explanatory efficacy experiences a discernible downturn. The average absolute value of the t-test statistic for interpreting its alpha is 2.075, notably surpassing the 5% significance threshold of 1.96. In this section, we examine the performance of both CPT3 and DS3 across different stages to elucidate the factors contributing to the subpar performance of CPT3 in the out-of-sample segment.

Firstly, it is noteworthy that, in contrast to Liu et al. (2022), the returns of MACP anomalies remain statistically significant even after undergoing explanation by CPT3, registering a weekly rate of 1.5% during the in-sample phase ($t=3.117$). Concurrently, a considerable decline is observed in the t-statistics of PRC and MAXPRC, dropping from 2.088 to 0.712 and from 2.353 to 0.816, respectively. This implies that the 30-40-30 grouping approach used for the CSMB factor in CPT3 does not fully capture market value anomalies. A

similar trend is evident across the entire sample period, where the returns of MACP, post-CPT3 explanation, achieve 1.0% ($t=2.971$). However, there is a significant reduction in the significance levels of PRCVOL and STDPRCVOL throughout the entire sample period. Furthermore, following Size-neutralization, CMOM shows a notable decrease in its explanatory capacity. In the in-sample phase, it falls short in explaining two-week momentum, with the alpha for two-week momentum at 1.6%. Despite a reduction in absolute magnitude, the t-value remains significant at 1.821 at the 10% significance level. Additionally, the CMOM factor does not exhibit a strong explanatory ability for two-week momentum in both the out-of-sample phase and the entire sample period.

We identify an additional concern in the CPT3 factor model, specifically its capacity to elevate initially insignificant anomalies to a significant level. Notably, in the realm of downside risk and tail risk, anomalies associated with Value-at-risk and expected-shortfall lacked significance before the CPT3 adjustment, with absolute value returns spanning from 0.07% to 1.6% (in-sample). However, following the CPT3 test, anomaly returns consistently achieved significance, with weekly returns surpassing 1.0%, and absolute t-values ranging from a minimum of 0.837 to a maximum of 3.103. Over the entire sample duration, the absolute values of downside risk and left-tail risk also manifested corresponding features.

To sum up, the lackluster performance of CPT3 in the out-of-sample and entire sample periods can be attributed to two main factors. Firstly, the factor model encounters difficulty in elucidating anomalies directly associated with the factor logic. This challenge may stem from differences in the grouping methodology, hinting at a potential resolution through adjustments to the grouping approach. Secondly, previously insignificant anomalies gain a level of significance after being interpreted by the factor model. Despite these anomalies falling outside the scope of those identified in Table 6, the emergence of noteworthy returns following factor model interpretation aligns with the statistical definition of anomalies.

[Insert Table 8 about here]

5. Robust Test

5.1 Fama-MacBeth regressions

To test the robustness of the single sorting test results, we employ the value-weighted Fama-MacBeth regression method. In accordance with Liu et al. (2022), we begin by sorting the features in ascending order and dividing the cryptocurrencies into five portfolios. We then replace the characteristic value in each portfolio with a corresponding rank from 1 to 5. This approach is useful in mitigating the impact of extreme values that are frequently observed in the cryptocurrency market, while also ensuring the relative size of the features

remains consistent. This methodology aids in analyzing the linear correlation between each characteristic and expected return, which enables us to evaluate the cross-sectional predictive power of each feature on expected returns.

We perform univariate FM regression

$$ExR_{i,t+1} = b_{0,t} + b_{1,t}X_{i,t} + \epsilon_{i,t+1},$$

where $ExR_{i,t+1}$ is the excess return, which is the difference between the weekly coin return on coin i and the “weekly” one-month T-bill rate at time $t + 1$, and $X_{i,t}$ is the independent variable. Considering that the returns of each portfolio in single sort is value-weighted, we also incorporate the influence of market capitalization in the FM regression. In other words, our regression in each period is actually a weighted least squares regression (WLS). In practice, we weight the residual sum of squares by the 0.5th power of market capitalization.

The findings indicate a general alignment between the WLS-FM regression outcomes and those of the Single Sorting test. Size-related factors like PRCVOL and STDPRCVOL demonstrate significance in both the in-sample and entire-sample periods but lose significance in the out-of-sample period. Notably, the t-statistics for MACP in the in-sample period (-2.516) and the out-of-sample period (-2.362) are both statistically significant at the 5% level. Conversely, the t-statistic for Size in the in-sample period is 0.04, and it lacks significance at the 5% level, consistent with the Single Sorting test results. Concerning Mom anomalies, various momentum measures exhibit significance at the 5% level across the three periods, highlighting a robust momentum effect in the cryptocurrency market. Similarly, Volume anomalies and those describing Volume volatility are significant in the in-sample and entire-sample periods but lose significance in the out-of-sample period.

However, the FM regression still presents anomalies that deviate from full consistency with the Single Sorting results. Notably, during the out-of-sample period, certain features within the Vol category, such as MAX, left-tail risk measures, and IDVOL, lack statistical significance. This suggests that the linear relationship between these features and expected returns is not statistically significant. Nevertheless, it is important to note that the regression coefficients for these features align with the sign of the average returns in Single Sorting, underscoring the robustness of our conclusions.

[Insert Table 9 about here]

5.2 Testing Factor Model Using Top 100 Coins

This subsection addresses the liquidity constraints associated with constructing investment portfolios in the cryptocurrency market, owing to the substantial differences in market capitalization among different cryptocurrencies. To address this issue, the subsection utilizes univariate sorting tests on the top 100 cryptocurrencies with the largest market capitalization for analysis at each time period. If the number of eligible cryptocurrencies in a given period is less than 100, all eligible cryptocurrencies are included for analysis. Table 10 presents the results of this single sorting test, using the top 100 cryptocurrencies by market capitalization.

[Insert Table 10 about here]

The findings indicate that the significance of MACP anomaly becomes weaker in the single sorting test due to smaller differences in market capitalization among the top 100 currencies compared to the full sample. Accordingly, the long-short combination of MACP still earns a significant average weekly return of -1.3%($t = -2.101$) during in-sample period and still shows statistically significance ($t = -1.847$) during full sample period at 10% significant level, as are the long-short return spreads on dollar trading volume and the standard deviation of dollar trading volume. However, momentum effects remain significant with a significant weekly return ranging within the three periods. Furthermore, we find that the RMOM anomalies shows great stability, earning weekly return at 2.4% during in-sample period, 1.7% during out-of-sample period and 2.2% during full sample period. The significant and weekly return of other significant anomalies like IDIOVOL during out-of-sample period or MAX during out-of-sample period decrease and become not significant at 5% anymore.

However, when we use the three factor models to explain the anomalies constructed with top 100 coins, our DS3 model still perform dominantly with the three sample periods.

[Insert Table 11 about here]

5.3 Excluding Effects of Grouping Method and Incorporate Insignificant Anomalies

In our dataset, the abnormal returns based on market value are significantly affected by the grouping method to the extent that factors constructed using the 30-40-30 grouping method are unable to account for the anomalies observed in the generated five groups. This disparity may be attributed to distinctions between our dataset and that of Liu et al (2022). Notably, this situation is objectively present, as illustrated by the

introduction of additional cryptocurrencies by CoinMarketCap post-July 2020, alongside the pre-existing presence of these currencies, causing differences between our dataset and Liu et al. (2022)’s dataset during the in-sample period.

Moreover, the aforementioned discussion takes into consideration that, following interpretation through the CPT3 model, some previously insignificant anomalies gradually become significant. However, in the prior analysis, we solely focused on the explanatory power of factors for significant anomalies, overlooking scenarios where alpha for insignificant anomalies becomes significant. From a statistical and economic standpoint, this suggests that the factor model falls short in explaining abnormal returns.

In this section, we concurrently address these two aspects. We substitute CSMB in CPT3 with the MACP anomaly from the anomalies and CMOM with $r_{3,0}$, testing the factor model’s explanatory power using all anomalies except for MACP and $r_{3,0}$. The results reveal that, even after adjusting the factor construction method and considering insignificant anomalies, our DS3 model continues to exhibit superiority in terms of mean absolute alpha and mean absolute t_alpha.

[Insert Table 12 about here]

6. Conclusion

In this investigation, we reassessed the performance of anomalies in the cryptocurrency market, drawing upon the research by Liu et al. (2022). Our examination covered their behavior across the in-sample, out-of-sample, and overall sample periods. Notably, we observed that while significant anomalies identified in the in-sample period largely retained their significance in the overall sample, the market value effect diminished in significance during the out-of-sample phase. Additionally, the predictive influence of left-tail risk on expected returns shifted from positive to negative, signaling substantial shifts in the cryptocurrency market during the out-of-sample period. This implies a potential need for a re-evaluation of factor models constructed based on earlier logic.

Moreover, by employing the Double Selection Lasso method, we crafted the cryptocurrency three-factor model (DS3) using in-sample data. Our DS3 model demonstrated a discernible advantage in elucidating anomalies in both the out-of-sample and overall sample periods when compared to Liu et al. (2022). We also compare the DS3 factor model with the IPCA3 factor model. The results show that while the IPCA3 model can reduce the scale of anomalies, the DS3 model demonstrates superior performance in eliminating the statistical significance of anomalies. Furthermore, factor spanning tests reveal that the DS3 model effectively incorporates information from both the CPT3 and IPCA3 models. We also conducted a concise analysis to

understand the reasons behind the underperformance of the CPT3 model. Lastly, our results underwent a battery of significance tests, encompassing stability assessments of different anomalies through WLS-FM regression, the creation of anomalies based on cryptocurrencies with large market capitalization to evaluate factor model explanatory power, and adjustments to the factor construction method of CPT3 to alleviate the impact of grouping on factor explanatory power, among other tests.

In summary, our findings affirm that the factor model constructed using DS-Lasso exhibits a certain superiority. From a historical standpoint, its explanatory capacity outperforms that of CPT3 and IPCA3.

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Table 1: Summary Statistics

The table reports the number of coins, the mean and the median of market capitalization, and the mean and median of daily trading price volume by year.

Year	#Coins	Market Cap (mil)		Volume (thous)	
		Mean	Median	Mean	Median
2014	121	232.58	3.81	1,230.47	32.95
2015	79	132.23	2.71	1,342.28	10.21
2016	157	157.32	3.31	1,837.42	20.62
2017	806	434.23	9.08	17,161.71	101.57
2018	1,569	360.57	8.89	18,841.20	96.71
2019	1,377	256.39	5.43	66,103.77	144.06
2020	1,636	352.56	6.71	127,191.27	271.00
2021	2,622	1,240.39	15.04	167,750.02	599.38
2022	2,149	911.67	13.41	110,526.31	506.69
2023	1,827	924.20	15.00	47,625.60	421.72
Total	4,193	742.57	10.11	92,475.90	305.86

Table 2: List of Anomalies

This table presents the anomaly we test in this paper, including their categories, name, reference, and definition. These anomalies are classified into seven categories, size, momentum (Mom), trading volume, volatility (Vol), downside risk, left-tail risk, and salience. The value-at-risk and expected shortfall are estimated by the non-parametric method.

Category	Anomaly	Reference	Definition
Size	<i>MACP</i>	Banz (1981)	Log Last day market capitalization in the portfolio formation week
	<i>PRC</i>	Miller and Scholes (1982)	Log last day price in the portfolio formation week
	<i>MAXPRC</i>	George and Hwang (2004)	The maximum price of the portfolio formation week
	<i>AGE</i>	Barry and Brown (1984)	The number of days that have been listed on Coinmarketcap.com
Mom	$r_{1,0}$	Jegadeesh and Titman (1993)	Past one-week return
	$r_{2,0}$	Jegadeesh and Titman (1993)	Past two-week return
	$r_{3,0}$	Jegadeesh and Titman (1993)	Past three-week return
	$r_{4,0}$	Jegadeesh and Titman (1993)	Past four-week return
	$r_{8,0}$	Jegadeesh and Titman (1993)	Past eight-week return
	$r_{3,1}$	Jegadeesh and Titman (1993)	Past one-to-three week return
	$r_{4,1}$	Jegadeesh and Titman (1993)	Past one-to-four-week return
	$r_{16,0}$	Jegadeesh and Titman (1993)	Past sixteen-week return
	$r_{50,0}$	De Bondt and Thaler (1985)	Past fifty-week return
	$r_{100,0}$	De Bondt and Thaler (1985)	Past one hundred-week return
	<i>CHMOM</i>	Gettleman and Marks (2006)	Past one-to-five-week return minus past six-to-ten week return
	<i>HIGH</i>	George and Hwang (2004)	The ratio of last day close price in the portfolio formation week to its highest close price during the past 365 days
	<i>RMOM</i>	Blitz, Huij, and Martens (2011)	The four-week residual momentum is the four-week cumulative residuals of regressing coin returns on the CAPM over the prior 4 weeks from week t-4 to week t
Volume	<i>PRCVOL</i>	Chordia, Subrahmanyam, and Anshuman (2001)	Log average dollar volume in the portfolio formation week
	<i>VOLSCALED</i>	Chordia, Subrahmanyam, and Anshuman (2001)	Average dollar volume scaled by market capitalization in the portfolio formation week
Vol	<i>BETA</i>	Fama and Macbeth (1973)	Estimation of beta in CAPM use daily returns of the previous 365 days before the portfolio formation week
	<i>BETA2</i>	Fama and Macbeth (1973)	Beta squared
	<i>IDIOVOL</i>	Ang, Hodrick, Xing, and Zhang (2006)	The idiosyncratic volatility is measured as the standard deviation of the residual after estimation CAPM. The model is estimated using daily returns of the previous 365 days before the formation week.
	<i>RETVOL</i>	Ang, Hodrick, Xing, and Zhang (2006)	The standard deviation of daily returns in the portfolio formation week.
	<i>MAX</i>	Bali, Cakici, and Whitelaw (2011)	Maximum daily return of the portfolio formation week

Table 2: List of Anomalies (Continued)

Category	Anomaly	Reference	Definition
Vol	<i>STDPRCVOL</i>	Chordia, Subrahmanyam, and Anshuman (2001)	Standard deviation of log daily dollar volume in the portfolio formation week
	<i>STDVOLSCALED</i>	Chordia, Subrahmanyam, and Anshuman (2001)	Standard deviation of dollar volume scaled by market capitalization in the portfolio formation week
	<i>AMIHUD</i>	Amihud (2002)	The average abs(daily return) divided by the price volume in the portfolio formation week
	<i>IDIOSKEW</i>	Ang, Hodrick, Xing, and Zhang (2006)	The idiosyncratic volatility is measured as the standard skewness of the residual after estimation CAPM. The model is estimated using daily returns of the previous 365 days before the formation week.
	<i>RETSKEW</i>	Amaya, Christoffersen, Jacobs and Vasquez (2015)	
	<i>COSKEW</i>	Harvey and Siddique (2000)	Co-skewness of coin return in relation to market risk in the portfolio formation week
	<i>DELAY</i>	Hou and Moskowitz (2005)	The difference of R squared between daily excess return regressed on the current, lagged one and lagged two day excess market return and the traditional CAPM model. The models are estimated using daily returns of the previous 365 days before the formation week.
	<i>DMBETA</i>	Dimson (1979)	We use the lead and lag of market returns along with the contemporaneous market returns, when estimating the market beta. The Dimson beta is calculated as the sum of the three coefficient of dependent variables. The model is estimated using daily returns of the previous 365 days before the formation week.
Downside Risk	<i>DOWNBETA</i>	Ang, Chen, and Xing (2006)	CAPM beta using returns if and only if the market return is less than average of market return during the regression period. The model is estimated using daily returns of the previous 365 days before the formation week.
	<i>SVAR</i>	Thierry and Pim (2004)	The Semi-Variance is calculated as the standard deviation of daily returns that are less than average during the past 365 days.
Left-Tail Risk	<i>VAR1</i>	Atilgan et al. (2020)	The 1% quantile of daily returns over the past 365 days multiply -1
	<i>ES1</i>	Atilgan et al. (2020)	The average of daily returns than are less than 1% quantile over the past 365 days multiply -1
	<i>VAR5</i>	Atilgan et al. (2020)	The 5% quantile of daily returns during the past 365 days multiply -1
	<i>ES5</i>	Atilgan et al. (2020)	The average of daily returns than are less than 5% quantile over the past 365 days multiply -1
	<i>VAR10</i>	Atilgan et al. (2020)	The 5% quantile of daily returns during the past 730 days multiply -1
	<i>ES10</i>	Atilgan et al. (2020)	The average of daily returns than are less than 5% quantile over the past 365 days multiply -1
	<i>MIN</i>	Bali, Cakici, and Whitelaw (2011)	Minimum daily return of the portfolio formation week multiply -1
	<i>DVAR1</i>	Sun, Wang, and Zhu (2022)	Change in VAR1
	<i>DES1</i>	Sun, Wang, and Zhu (2022)	Change in ES1
	<i>DVAR5</i>	Sun, Wang, and Zhu (2022)	Change in VAR5
	<i>DES5</i>	Sun, Wang, and Zhu (2022)	Change in ES5
	<i>DVAR10</i>	Sun, Wang, and Zhu (2022)	Change in VAR10
	<i>DES10</i>	Sun, Wang, and Zhu (2022)	Change in ES10
Salience	<i>STR</i>	Cosemans and Frehen (2021)	Salience theory of daily return during the formation week
	<i>STV</i>	Sun, Wang, and Zhu (2023)	Salience theory of turnover during the formation week

Table 3: Single Sort

This table presents the value-weighted averages of weekly cryptocurrency returns, along with their corresponding t -values, for long-short quintile portfolios organized based on prior-week characteristics as outlined in Table 2. In Column (1), the results are derived using data consistent with Liu et al. (2022) spanning from January 2014 to July 2020. Column (2) displays results from August 2020 to December 2023. The data used in Column (3) covers the period from January 2014 to December 2023. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Category	Anomaly	(1)		(2)		(3)	
		return	t-value	return	t-value	return	t-value
Size	<i>MACP</i>	-0.025**	-3.711	-0.001	-0.296	-0.017	-3.617
	<i>PRC</i>	-0.024**	-2.088	-0.009	-1.116	-0.019**	-2.340
	<i>MAXPRC</i>	-0.031**	-2.353	-0.007	-0.954	-0.023**	-2.528
	<i>AGE</i>	0.006	0.723	0.002	0.421	0.005	0.809
Mom	$r_{1,0}$	0.036***	2.635	0.016**	2.317	0.029***	3.157
	$r_{2,0}$	0.040***	3.779	0.038***	4.943	0.039***	5.322
	$r_{3,0}$	0.040***	3.831	0.022***	3.026	0.033***	4.662
	$r_{4,0}$	0.031***	3.065	0.020***	2.743	0.027***	3.871
	$r_{8,0}$	0.015	1.297	0.007	1.000	0.012	1.561
	$r_{3,1}$	0.026**	2.440	0.021***	2.963	0.024***	3.316
	$r_{4,1}$	0.018*	1.923	0.013**	1.960	0.017**	2.511
	$r_{16,0}$	0.007	0.700	0.007	1.082	0.007	1.036
	$r_{50,0}$	-0.010	-1.229	0.009	1.548	-0.003	-0.507
	$r_{100,0}$	-0.005	-0.450	0.002	0.361	-0.002	-0.278
	<i>CHMOM</i>	0.006	0.580	0.010	1.268	0.007	1.044
	<i>HIGH</i>	-0.006	-0.581	-0.005	-0.630	-0.005	-0.805
	<i>RMOM</i>	0.021**	2.422	0.012**	2.378	0.018***	3.046
Volume	<i>PRCVOL</i>	-0.024**	-2.495	0.003	0.675	-0.015**	-2.252
	<i>VOLSCALED</i>	-0.014	-0.875	0.001	0.164	-0.008	-0.807
Vol	<i>BETA</i>	-0.012	-1.254	0.014	1.577	-0.002	-0.270
	<i>BETA2</i>	-0.014	-1.440	0.014	1.515	-0.003	-0.456
	<i>IDIOVOL</i>	0.014	1.372	-0.015**	-1.953	0.004	0.527
	<i>RETVOL</i>	0.002	0.163	-0.005	-0.527	-0.001	-0.086
	<i>MAX</i>	0.017	1.214	-0.010	-1.358	0.007	0.787
	<i>STDPRCVOL</i>	-0.027***	-3.025	-0.001	-0.106	-0.018***	-2.897
	<i>STDVOLSCALED</i>	-0.009	-0.917	-0.003	-0.510	-0.007	-1.038
	<i>AMIHU</i>	0.021***	2.577	0.000	0.027	0.014**	2.407
	<i>IDIOSKEW</i>	0.003	0.250	-0.007	-1.119	-0.001	-0.177
	<i>RETSKEW</i>	0.004	0.364	-0.008	-1.444	-0.001	-0.121
	<i>COSKEW</i>	-0.001	-0.132	-0.006	-1.068	-0.003	-0.621
	<i>DELAY</i>	-0.002	-0.227	-0.007	-1.109	-0.004	-0.730
	<i>DMBETA</i>	-0.007	-0.626	0.009	1.034	0.000	-0.063
Downside Risk	<i>DOWNBETA</i>	-0.009	-0.937	0.016*	1.806	0.000	0.055
	<i>SVAR</i>	0.004	0.254	-0.010*	-1.858	-0.002	-0.202
Left-tail Risk	<i>VAR1</i>	0.016	1.276	-0.009	-1.343	0.007	0.801
	<i>ES1</i>	0.007	0.699	-0.010*	-1.663	0.000	0.044
	<i>VAR5</i>	0.014	1.326	-0.007	-0.972	0.006	0.845
	<i>ES5</i>	0.010	0.965	-0.012*	-1.824	0.001	0.175
	<i>VAR10</i>	0.013	1.233	-0.011*	-1.641	0.004	0.526
	<i>ES10</i>	0.012	1.159	-0.011	-1.562	0.003	0.460
	<i>MIN</i>	-0.002	-0.146	-0.025***	-3.990	-0.010	-1.019
	<i>DVAR1</i>	0.002	0.190	0.013**	2.265	0.006	1.006
	<i>DES1</i>	0.007	0.566	0.013**	2.281	0.009	1.150
	<i>DVAR5</i>	-0.011	-1.284	0.007	1.263	-0.004	-0.676
	<i>DES5</i>	-0.005	-0.665	0.006	1.002	-0.001	-0.165
	<i>DVAR10</i>	-0.008	-1.014	0.006	1.052	-0.003	-0.462
	<i>DES10</i>	-0.007	-0.716	0.001	0.193	-0.004	-0.589
Salience	<i>STR</i>	0.007	0.508	0.005	0.882	0.007	0.690
	<i>STV</i>	0.000	-0.041	-0.003	-0.565	-0.001	-0.224

Table 4: Single Sort: Long-term test (2014.1.1 – 2023.12.31)

Table 4 presents the value-weighted average weekly returns as well as their t -values in T+k week according to 5-1 portfolio formed at the end of week T in table 2. Panel A shows the significant anomalies and panel B shows the insignificant anomalies. The t -values with bold font denote statistical significance at the 5% level.

Panel A : Significant Anomalies											
Category	Anomaly	T+2		T+3		T+4		T+5		T+6	
		Excess	t -value	Excess	t -value	Excess	t -value	Excess	t -value	Excess	t -value
Size	MACP	-0.020	-3.189	-0.024	-3.818	-0.027	-3.898	-0.021	-4.113	-0.021	-4.290
	PRC	-0.020	-2.696	-0.018	-2.429	-0.013	-1.910	-0.019	-2.894	-0.017	-2.510
	MAXPRC	-0.019	-2.493	-0.019	-2.516	-0.013	-1.971	-0.017	-2.534	-0.018	-2.623
Mom	$r_{1,0}$	0.025	3.757	0.014	1.886	0.007	0.981	-0.010	-1.526	-0.002	-0.206
	$r_{2,0}$	0.026	3.634	0.002	0.242	-0.011	-1.250	-0.014	-1.745	0.000	-0.048
	$r_{3,0}$	0.016	2.554	-0.006	-0.815	-0.019	-2.338	-0.005	-0.772	0.001	0.076
	$r_{4,0}$	0.003	0.436	-0.010	-1.261	-0.003	-0.506	-0.008	-1.227	-0.003	-0.315
	$r_{3,1}$	0.003	0.388	-0.010	-1.082	-0.019	-2.469	-0.001	-0.134	0.005	0.707
	$r_{4,1}$	-0.005	-0.652	-0.016	-1.933	-0.010	-1.474	-0.002	-0.180	-0.004	-0.439
	RMOM	0.007	1.208	0.005	0.851	-0.003	-0.638	-0.006	-1.129	0.002	0.352
Volume	PRCVOL	-0.025	-4.243	-0.025	-4.402	-0.024	-3.995	-0.029	-2.744	-0.028	-3.229
	VOLSCALED	-0.014	-1.385	-0.013	-1.367	-0.025	-2.189	-0.024	-1.973	-0.019	-1.838
Vol	STDPRCVOL	-0.031	-2.791	-0.026	-4.338	-0.035	-2.962	-0.035	-2.653	-0.028	-3.229
	AMIHU	0.025	3.804	0.023	3.868	0.024	3.921	0.025	2.370	0.028	2.650
	STDVOLSCALED	-0.009	-1.101	-0.009	-1.241	-0.027	-2.625	-0.016	-2.049	-0.015	-1.599
Left-Tail Risk	VAR1	0.004	0.698	0.004	0.706	0.009	1.669	0.011	2.046	0.012	2.009
	ES1	0.008	1.194	0.007	1.367	0.010	1.783	0.011	1.846	0.012	1.957
	VAR5	0.008	1.272	0.007	1.237	0.011	1.848	0.010	1.808	0.012	2.077
	ES5	0.007	1.067	0.008	1.500	0.010	1.753	0.010	1.973	0.011	1.861
	VAR10	0.009	1.361	0.007	1.181	0.013	2.147	0.012	2.214	0.014	2.319
	ES10	0.008	1.232	0.005	0.952	0.009	1.700	0.010	1.788	0.011	1.881

Table 4 (Continued)

Panel B : Insignificant Anomalies											
Category	Anomaly	T+2		T+3		T+4		T+5		T+6	
		Excess	<i>t</i> -value	Excess	<i>t</i> -value	Excess	<i>t</i> -value	Excess	<i>t</i> -value	Excess	<i>t</i> -value
Size	AGE	0.002	0.321	0.001	0.097	0.001	0.170	-0.001	-0.109	-0.001	-0.144
Mom	$r_{8,0}$	0.001	0.117	-0.019	-1.762	-0.012	-1.298	-0.007	-1.120	-0.009	-1.631
	$r_{16,0}$	-0.002	-0.330	-0.007	-1.019	-0.012	-1.846	-0.012	-2.004	-0.008	-1.388
	$r_{50,0}$	-0.003	-0.469	-0.004	-0.698	-0.007	-1.051	-0.011	-1.987	-0.007	-1.363
	$r_{100,0}$	0.007	1.019	-0.003	-0.520	-0.002	-0.307	-0.002	-0.366	-0.002	-0.252
	<i>CHMOM</i>	0.000	0.062	-0.001	-0.172	-0.005	-0.872	-0.005	-0.939	-0.008	-1.594
	HIGH	0.001	0.117	-0.002	-0.341	-0.002	-0.325	-0.002	-0.314	-0.007	-1.057
Vol	<i>BETA</i>	-0.001	-0.095	-0.001	-0.126	0.001	0.129	0.001	0.081	0.003	0.438
	<i>BETA2</i>	-0.001	-0.186	0.000	0.008	0.002	0.350	0.001	0.208	0.003	0.435
	IDIOVOL	0.010	1.340	0.000	-0.028	0.006	0.663	-0.001	-0.140	-0.008	-1.392
	RETVOL	0.002	0.288	-0.002	-0.317	0.001	0.159	0.003	0.515	0.006	0.981
	MAX	0.007	1.080	-0.003	-0.409	0.001	0.089	-0.005	-0.783	-0.004	-0.617
	IDIOSKEW	0.004	0.587	0.004	0.573	0.006	0.769	0.004	0.587	0.006	0.776
	RETSKEW	0.003	0.443	0.001	0.112	0.001	0.160	-0.003	-0.411	-0.002	-0.264
	COSKEW	-0.007	-1.269	-0.009	-1.686	-0.005	-1.048	-0.002	-0.421	0.000	0.069
	DELAY	-0.005	-0.904	-0.004	-0.833	-0.001	-0.287	-0.001	-0.144	-0.003	-0.694
	DMBETA	-0.004	-0.511	-0.001	-0.162	0.000	0.061	0.001	0.109	0.001	0.186
Downside Risk	<i>DOWNBETA</i>	0.001	0.098	0.001	0.156	0.004	0.632	0.001	0.090	0.000	-0.032
	<i>SVAR</i>	-0.001	-0.137	-0.002	-0.393	0.004	0.604	0.006	1.023	0.007	1.101
Left-tail Risk	MIN	-0.007	-1.042	-0.003	-0.427	-0.004	-0.550	0.008	0.789	0.005	0.610
	DVAR1	-0.007	-1.165	0.004	0.750	-0.014	-2.082	-0.008	-1.228	-0.015	-2.296
	DES1	-0.008	-1.339	0.004	0.829	-0.011	-1.582	-0.007	-1.147	-0.015	-2.172
	DVAR5	0.005	0.873	-0.001	-0.165	-0.003	-0.590	0.010	2.015	-0.004	-0.805
	DES5	-0.002	-0.322	-0.003	-0.571	-0.003	-0.532	0.003	0.634	-0.009	-1.868
	DVAR10	-0.009	-1.606	0.006	1.218	0.004	0.734	0.004	0.700	-0.003	-0.702
	DES10	-0.006	-0.997	-0.003	-0.560	-0.008	-1.296	-0.002	-0.305	-0.004	-0.699
Salience	STR	-0.003	-0.421	0.002	0.285	-0.003	-0.345	-0.006	-0.794	0.000	-0.031
	STV	0.003	0.449	0.011	1.983	-0.004	-0.898	-0.009	-1.453	0.002	0.476

Table 5: Descriptive Statistics of Factors

This table primarily presents the average factor returns, t -values, and correlations between factors within the DS3 model before orthogonalization, the DS3 model after orthogonalization, Liu et al.'s (2022) CPT3 model and IPCA3 model. MKT represents the market value-weighted returns of all cryptocurrencies in the sample; MOM2 denotes the two-week momentum factor; Raw RMOM is the residual momentum factor, calculated as the four-week cumulative residuals of regressing coin returns on the CAPM over the prior 4 weeks from week $t-4$ to week t ; RMOM represents the orthogonalized residual momentum factor. CSMB and CMOM are the market value and momentum factors, respectively, in Liu et al.'s (2022) model. IPCA_1, IPCA_2, and IPCA_3 represent the three factors in the IPCA3 model. The sample period is from January 2014 to December 2023.

	Return	T-value	Correlation								
			MKT	MOM2	RMOM	Raw RMOM	CSMB	CMOM	IPCA_1	IPCA_2	IPCA_3
MKT	0.012	2.652	1.00								
MOM2	0.031	5.541	0.10	1.00							
RMOM	0.014	3.720	0.04	0.16	1.00						
Raw RMOM	0.018	3.833	0.02	0.35	0.72	1.00					
CSMB	0.015	2.954	0.01	0.24	0.19	0.27	1.00				
CMOM	0.018	4.280	0.05	0.62	0.24	0.32	0.21	1.00			
IPCA_1	1.840	1.475	0.07	0.14	0.18	0.06	0.12	0.07	1.00		
IPCA_2	0.072	2.017	0.18	0.28	0.22	0.21	0.50	0.23	-0.01	1.00	
IPCA_3	0.044	2.421	-0.05	-0.03	0.09	0.07	-0.00	0.28	-0.01	-0.01	1.00

Table 6: Comparison of Factor Models

This table illustrates the explanatory power of the CAPM, CPT3, DS3 and IPCA3 models on anomalies. Panel A presents the explanatory capabilities of the models for the in-sample period from January 2014 to July 2020. Panel B shows the out-of-sample explanatory power of the factor models for the period from August 2020 to December 2023. Panel C shows the full sample period explanatory power of the factor models for the period. Throughout the three observation periods, only the significant anomalies within the 5% threshold will be considered in the examination of the explanatory power of factor models. The comparative indicators includes the average absolute alpha after adjustment, the average absolute alphas' t -statistics and the p -value of Gibbons–Ross–Shanken (1989) F-test on those anomalies, the number of minimum alphas which is the number of anomalies for which the factor model produces the smallest absolute alpha among the models being compared in the table and the number of unexplained alphas which is the number of anomalies with t -values of the factor models' alphas greater than 1.96 .

Panel A: In-sample Test				
	CAPM	CPT3	DS3	IPCA3
Avg_abs_alpha (%)	2.727	0.972	1.011	0.120
Avg_abs_t_alpha	2.785	1.291	1.317	3.726
GRS p-value	0.001	0.131	0.126	0.008
num_min_abs_alpha	0	0	1	11
Num_unexplained_alpha	11	1	2	9
Panel B: Out-of-sample Test				
	CAPM	CPT3	DS3	IPCA3
Avg_abs_alpha (%)	1.950	1.146	0.592	2.036
Avg_abs_t_alpha	3.056	2.075	1.148	0.800
GRS p-value	0.000	0.000	0.027	0.000
num_min_abs_alpha	0	0	9	0
Num_unexplained_alpha	7	2	1	0
Panel C: Full Sample Period				
	CAPM	CPT3	DS3	IPCA3
Avg_abs_alpha (%)	2.155	0.814	0.741	0.118
Avg_abs_t_alpha	3.213	1.560	1.338	3.529
GRS p-value	0.000	0.008	0.061	0.000
num_min_abs_alpha	0	2	2	9
Num_unexplained_alpha	12	3	2	9

Table 7: Abilities to explain each other's factor

Panel A reports a factor's estimated weekly alpha (in percent) with respect to the other model. The t -values are adjusted by Newey-West Standard Error. Panel B reports the F -statistic and associated p -value in the "GRS" test of Gibbons–Ross–Shanken (1989) on the null hypothesis that a given model produces zero alphas for the factors of the other model. The sample period is January 2014 to December 2023. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Factors	Alphas(%) with respect to		
Panel A: alphas	CPT3	DS3	IPCA3
MKT	-		1.051** (2.097)
CSMB	-	0.712 (1.166)	0.911* (1.728)
CMOM	-	0.075 (0.207)	1.156** (3.220)
RMOM	0.894*** (2.579)	-	0.951** (2.471)
MOM2	1.476*** (3.482)	-	2.553** (2.553)
IPCA_1	80.490 (0.838)	4.697 (0.103)	
IPCA_2	0.620 (0.287)	1.742 (0.688)	
IPCA_3	2.035 (1.258)	3.564** (1.989)	
Panel B: GRS P-value			
CSMB, CMOM		0.897 (0.408)	6.765 (0.001)
RMOM, MOM2	7.883*** (0.000)		16.048*** (0.000)
IPCA_1, IPCA_2, IPCA_3	1.085 (0.355)	1.841 (0.139)	

Table 8: Anomalies adjusted by CPT3

This table presents the alpha of anomalies after adjusted by CPT3 model introduced by Liu et al. (2022), along with their corresponding t -values. Before conducting regression analysis, we standardize all variables to be positively oriented. Specifically, for factors with t -values less than 0 during the full-sample period, we take the opposite of their original return rates. In Column (1), the results are derived using data consistent with Liu et al. (2022) spanning from January 2014 to July 2020. Column (2) displays results from August 2020 to December 2023. The data used in Column (3) covers the period from January 2014 to December 2023. The reported t -statistics in this table are calculated using Newey-West robust standard errors. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Category	Anomaly	(1)		(2)		(3)	
		return	t-value	return	t-value	return	t-value
Size	<i>MACP</i>	0.015***	3.117	0.000	0.295	0.010***	2.971
	<i>PRC</i>	0.007	0.712	0.005	0.993	0.006	0.875
	<i>MAXPRC</i>	0.008	0.816	0.003	0.779	0.006	0.902
	<i>AGE</i>	0.001	0.140	0.002	0.454	0.001	0.244
Mom	$r_{1,0}$	0.018	1.492	0.008	1.347	0.015*	1.869
	$r_{2,0}$	0.016*	1.821	0.023***	4.071	0.020***	3.288
	$r_{3,0}$	0.010*	1.725	0.007	1.457	0.010**	2.447
	$r_{4,0}$	0.006	0.832	0.008	1.497	0.008	1.545
	$r_{8,0}$	-0.006	-0.555	-0.002	-0.326	-0.004	-0.660
	$r_{3,1}$	0.004	0.543	0.009	1.441	0.007	1.263
	$r_{4,1}$	-0.004	-0.523	0.004	0.785	0.000	-0.041
	$r_{16,0}$	-0.010	-1.140	0.000	0.019	-0.006	-1.020
	$r_{50,0}$	0.011	1.572	-0.004	-0.704	0.005	1.009
	$r_{100,0}$	0.009	1.138	-0.001	-0.284	0.005	0.973
	<i>CHMOM</i>	-0.012	-1.498	0.006	1.099	-0.006	-1.046
	<i>HIGH</i>	0.003	0.406	-0.001	-0.113	0.003	0.541
	<i>RMOM</i>	0.009	1.166	0.005	1.224	0.009	1.722
Volume	<i>PRCVOL</i>	0.004	0.471	0.001	0.151	0.002	0.394
	<i>VOLSCALED</i>	0.011	0.726	0.001	0.118	0.008	0.755
Vol	<i>BETA</i>	0.012	1.312	0.000	0.087	0.007	1.036
	<i>BETA2</i>	0.014	1.488	0.001	0.232	0.008	1.244
	<i>IDIOVOL</i>	-0.011	-1.351	-0.019***	-3.269	-0.014**	-2.458
	<i>RETVOL</i>	0.013	1.338	0.008	0.972	0.010	1.420
	<i>MAX</i>	-0.008	-0.706	-0.018***	-3.706	-0.012	-1.635
	<i>STDPRCVOL</i>	0.010	1.354	0.004	1.001	0.008	1.504
	<i>STDVOLSCALED</i>	0.000	0.049	0.003	0.592	0.002	0.361
	<i>AMIHU</i>	0.010	1.336	0.003	0.613	0.007	1.320
	<i>IDIOSKEW</i>	0.008	1.089	0.008	1.439	0.008	1.487
	<i>RETSKEW</i>	0.007	0.877	0.011**	2.003	0.008	1.392
	<i>COSKEW</i>	0.000	-0.043	0.004	0.815	0.002	0.486
	<i>DELAY</i>	0.011	1.504	0.004	0.932	0.006	0.986
	<i>DMBETA</i>	0.009	0.855	0.003	0.562	0.008	1.565
Downside Risk	<i>DOWNBETA</i>	-0.013*	-1.822	0.002	0.532	-0.007	-1.306
	<i>SVAR</i>	0.000	0.034	0.009*	1.954	0.003	0.341
Left-tail Risk	<i>VAR1</i>	-0.009	-0.837	-0.006	-1.074	-0.008	-1.031
	<i>ES1</i>	-0.021***	-3.103	-0.007	-1.278	-0.016***	-3.467
	<i>VAR5</i>	-0.015**	-2.059	-0.004	-0.622	-0.011**	-2.245
	<i>ES5</i>	-0.018***	-2.737	-0.009	-1.567	-0.015***	-3.254
	<i>VAR10</i>	-0.017***	-2.576	-0.010	-1.570	-0.015***	-3.203
	<i>ES10</i>	-0.016**	-2.351	-0.008	-1.309	-0.014***	-2.825
	<i>MIN</i>	0.009	0.692	0.026***	4.118	0.016 *	1.894
	<i>DVAR1</i>	0.002	0.261	0.008	1.573	0.005	0.810
	<i>DES1</i>	0.008	0.616	0.009	1.609	0.008	1.036
	<i>DVAR5</i>	0.005	0.639	-0.002	-0.322	0.001	0.260
	<i>DES5</i>	0.002	0.342	-0.004	-0.965	0.000	-0.002
	<i>DVAR10</i>	0.003	0.416	-0.006	-1.108	-0.001	-0.147
	<i>DES10</i>	0.003	0.479	-0.001	-0.143	0.002	0.403
Salience	<i>STR</i>	0.001	0.057	0.001	0.278	0.001	0.185
	<i>STV</i>	-0.002	-0.258	0.001	0.286	-0.001	-0.148

Table 9: t -value of slope in Fama-Macbeth Regression

This table shows the t -value of slope in univariate value-weighted Fama-MacBeth Regression during in sample period (1), out-of-sample period (2) and full sample period (3). Following Liu et al. (2022), we first sort the features in ascending order and divide the cryptocurrencies into five portfolios, replacing the characteristic value in each portfolio with 1 to 5. We use weighted least square to estimate the equation. The t -statistics reported in this table are based on Newey-West robust standard error. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Category	Anomaly	(1)	(2)	(3)
Size	MACP	-2.516**	0.040	-2.362**
	PRC	-2.189**	-0.797	-2.315**
	MAXPRC	-2.190**	-0.684	-2.279**
	AGE	-1.098	2.121**	-0.573
Mom	$r_{1,0}$	3.349***	2.475**	3.896***
	$r_{2,0}$	4.438***	4.476***	5.618***
	$r_{3,0}$	4.201***	3.197***	4.919***
	$r_{4,0}$	3.813***	3.677***	4.765***
	$r_{8,0}$	1.307	1.627	1.758*
	$r_{3,1}$	3.532***	2.739***	4.091***
	$r_{4,1}$	3.389***	3.591***	4.322***
	$r_{16,0}$	0.732	0.308	0.792
	$r_{50,0}$	-1.642	1.066	-1.093
	$r_{100,0}$	-0.906	-0.989	-1.259
	CHMOM	1.553	1.588	2.066***
	HIGH	-0.885	1.164	-0.312
	RMOM	3.804***	2.547**	4.320***
Volume	PRCVOL	-2.671***	0.162	-2.544**
	VOLSCALED	-2.351**	-0.417	-2.354**
Vol	BETA	-1.324	0.748	-0.87
	BETA2	-1.029	0.880	-0.542
	IDIOVOL	1.212	-0.894	0.860
	RETVOL	0.12	-0.334	-0.016
	MAX	2.110**	0.153	2.048**
	STDPRCVOL	-2.740***	-0.096	-2.662***
	STDVOLSCALED	-2.552***	-1.006	-2.701***
	AMIHU	1.940*	-0.424	1.762*
	IDIOSKEW	0.709	-0.988	0.335
	RETSKEW	0.360	-1.877*	-0.258
	COSKEW	-1.250	-0.967	-1.561
	DELAY	0.674	-0.709	0.396
	DMBETA	-0.553	0.534	-0.284
Downside Risk	DOWNBETA	-0.267	0.936	0.195
	SVAR	0.039	-0.201	-0.059
Left-Tail Risk	VAR1	1.295	-0.265	1.148
	ES1	1.332	-0.113	1.216
	VAR5	1.447	0.209	1.424
	ES5	1.243	0.000	1.169
	VAR10	1.415	0.297	1.420
	ES10	1.327	-0.029	1.233
	MIN	0.813	-2.133**	0.192
	DVAR1	-1.011	2.486**	-0.246
	DES1	-0.853	2.194**	-0.159
	DVAR5	-0.203	2.398**	0.728
	DES5	-0.909	2.129**	-0.02
	DVAR10	-1.338	1.620	-0.769
	DES10	-1.175	1.804*	-0.443
Salience	STR	0.962	0.735	1.075
	STV	0.632	-1.308	0.291

Table 10: Single Sort: top 100 coins only

This table presents the value-weighted averages of weekly returns and the CAPM alphas as well as their t -values, for long-short quintile portfolio sorted by characteristics, showed in Table 2, of top 100 coins. We only show the results which we focus on. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)		(2)		(3)	
Anomaly	return	t-value	return	t-value	return	t-value
<i>MACP</i>	-0.013**	-2.101	0.001	0.235	-0.008*	-1.847
<i>PRC</i>	-0.019*	-1.904	-0.006	-1.082	-0.014**	-2.137
<i>MAXPRC</i>	-0.023**	-2.017	-0.006	-1.136	-0.017**	-2.242
$r_{1,0}$	0.033**	2.473	0.016**	2.231	0.028***	3.003
$r_{2,0}$	0.037***	3.354	0.026***	3.565	0.033***	4.381
$r_{3,0}$	0.042***	3.942	0.020***	3.112	0.034***	4.706
$r_{4,0}$	0.030***	2.832	0.020***	2.748	0.027***	3.625
$r_{3,1}$	0.031***	2.865	0.014**	2.102	0.025***	3.369
$r_{4,1}$	0.018*	1.897	0.013*	1.916	0.016**	2.483
<i>RMOM</i>	0.024***	2.798	0.017***	2.776	0.022***	3.608
<i>PRCVOL</i>	-0.018*	-1.826	0.003	0.429	-0.011	-1.594
<i>VOLSCALED</i>	-0.009	-0.535	0.006	0.664	-0.004	-0.327
<i>IDIOVOL</i>	0.012	1.141	-0.010	-1.430	0.004	0.599
<i>RETVOL</i>	-0.008	-0.973	0.000	-0.018	-0.005	-0.872
<i>MAX</i>	0.016	1.124	-0.007	-1.099	0.008	0.844
<i>STDPRCVOL</i>	-0.018**	-2.016	0.000	-0.002	-0.012*	-1.898
<i>STDVOLSCALED</i>	-0.005	-0.532	0.008	0.965	-0.001	-0.075
<i>AMIHUD</i>	0.011	1.470	-0.006	-0.980	0.005	0.907
<i>DOWNBETA</i>	-0.004	-0.426	0.004	0.576	-0.001	-0.181
<i>SVAR</i>	-0.004	-0.442	0.001	0.172	-0.002	-0.328
<i>VAR1</i>	-0.001	-0.122	-0.006	-0.987	-0.003	-0.486
<i>ES1</i>	0.002	0.166	0.000	-0.064	0.001	0.126
<i>VAR5</i>	0.010	0.927	0.005	0.697	0.009	1.131
<i>ES5</i>	0.000	-0.020	-0.003	-0.458	-0.001	-0.207
<i>VAR10</i>	0.006	0.547	-0.001	-0.087	0.003	0.468
<i>ES10</i>	0.003	0.314	-0.001	-0.186	0.002	0.216
<i>MIN</i>	-0.008	-0.490	-0.017**	-2.508	-0.011	-1.053

Table 11: Comparison of Factor Models with anomalies constructed by top 100

This table illustrates the explanatory power of the CAPM, CPT3, DS3 and IPCA3 models on anomalies constructed by top 100 coins. Panels A, B, and C presents the explanatory capabilities of the models for the in-sample period from January 2014 to July 2020, out-sample period from August 2020 to December 2023, and full sample period from January 2014 to December 2023, respectively. We separately exam the comparative indicators includes the average absolute alpha after adjustment, the average absolute alphas' t -statistics and the p -value of Gibbons–Ross–Shanken (1989) F -test on those anomalies, the number of minimum alphas which is the number of anomalies for which the factor model produces the smallest absolute alpha among the models being compared in the table and the number of unexplained alphas which is the number of anomalies with t -values of the factor models' alphas greater than 1.96 .

Panel A: In-sample Test				
	CAPM	CPT3	DS3	IPCA
Avg_abs_alpha (%)	2.693	0.917	0.705	0.124
Avg_abs_t_alpha	2.785	1.165	0.953	4.349
GRS p-value	0.003	0.389	0.414	0.027
num_min_abs_alpha	1	1	0	8
Num_unexplained_alpha	7	0	1	5
Panel B: Out-of-sample Test				
	CAPM	CPT3	DS3	IPCA
Avg_abs_alpha (%)	1.802	0.981	0.336	2.966
Avg_abs_t_alpha	2.884	1.847	0.639	0.919
GRS p-value	0.004	0.045	0.580	0.001
num_min_abs_alpha	0	0	8	0
Num_unexplained_alpha	7	3	0	0
Panel B: Full Sample Period				
	CAPM	CPT3	DS3	IPCA
Avg_abs_alpha (%)	2.275	0.854	0.439	0.116
Avg_abs_t_alpha	3.385	1.517	0.736	4.234
GRS p-value	0.000	0.189	0.649	0.002
num_min_abs_alpha	0	1	4	5
Num_unexplained_alpha	7	4	0	5

Table 12: Comparison of Factor Models with Different Grouping Methods for all anomalies
This table illustrates the explanatory power of the CAPM, CPT3, DS3 and IPCA3 models for all anomalies. Here, we use MACP instead of size to construct the CSMB factor and anomaly $r_{3,0}$ to construct CMOM factor for the CPT3 model. Panels A, B, and C presents the explanatory capabilities of the models for the in-sample period from January 2014 to July 2020, out-sample period from August 2020 to December 2023, and full sample period from January 2014 to December 2023 respectively. We separately examine the explanatory abilities of the three models in these two intervals concerning all anomalies except MACP and $r_{3,0}$, despite of significance. The comparative indicators includes the average absolute alpha after adjustment, the average absolute alphas' t -statistics and the p -value of Gibbons–Ross–Shanken (1989) F -test on those anomalies, the number of minimum alphas which is the number of anomalies for which the factor model produces the smallest absolute alpha among the models being compared in the table and the number of unexplained alphas which is the number of anomalies with t -values of the factor models' alphas greater than 1.96 .

Panel A: In-sample Test				
	CAPM	CPT3	DS3	IPCA
Avg_abs_alpha (%)	1.282	0.835	0.762	0.245
Avg_abs_t_alpha	1.303	1.207	0.933	4.678
GRS p-value	0.002	0.000	0.044	0.001
num_min_abs_alpha	5	4	6	34
Num unexplained alpha	11	9	4	27
Panel B: Out-of-sample Test				
	CAPM	CPT3	DS3	IPCA
Avg_abs_alpha (%)	0.771	0.530	0.426	3.398
Avg_abs_t_alpha	1.279	1.236	0.754	1.143
GRS p-value	0.000	0.000	0.003	0.000
num_min_abs_alpha	9	11	25	4
Num unexplained alpha	11	8	1	6
Panel C: Full Sample Period				
	CAPM	CPT3	DS3	IPCA
Avg_abs_alpha (%)	0.942	0.942	0.610	0.258
Avg_abs_t_alpha	1.497	1.497	0.973	4.847
GRS p-value	0.000	0.000	0.000	0.000
num_min_abs_alpha	7	5	9	28
Num unexplained alpha	13	13	3	26

Appendix

Table A1 Comparison of Factor Models Using CryptoCompare Data

This table illustrates the explanatory power of the CAPM, CPT3, DS3 and IPCA3 models on anomalies, with all analyses conducted using data from CryptoCompare. Panel A presents the explanatory capabilities of the models for the in-sample period from March 2018³ to July 2020. Panel B shows the out-of-sample explanatory power of the factor models for the period from August 2020 to December 2023. Panel C shows the full sample period explanatory power of the factor models for the period. Throughout the three observation periods, only the significant anomalies within the 5% threshold will be considered in the examination of the explanatory power of factor models. The comparative indicators include the average absolute alpha after adjustment, the average absolute alphas' t-statistics and the p-value of Gibbons–Ross–Shanken (1989) F-test on those anomalies, the number of minimum alphas which is the number of anomalies for which the factor model produces the smallest absolute alpha among the models being compared in the table and the number of unexplained alphas which is the number of anomalies with t-values of the factor models' alphas greater than 1.96.

Panel A: In-sample Test				
	CAPM	CPT3	DS3	IPCA3
Avg_abs_alpha (%)	1.489	1.262	1.252	4.634
Avg_abs_t_alpha	2.154	2.531	1.987	2.104
GRS p-value	0.02	0.00	0.06	0.06
num_min_abs_alpha	0	0	2	0
Num unexplained alpha	1	2	1	1
Panel B: Out-of-sample Test				
	CAPM	CPT3	DS3	IPCA3
Avg_abs_alpha (%)	1.818	1.124	0.737	1.993
Avg_abs_t_alpha	2.829	2.052	1.297	0.707
GRS p-value	0.00	0.00	0.05	0.00
num_min_abs_alpha	1	0	5	1
Num unexplained alpha	7	3	2	0
Panel C: Full Sample Period				
	CAPM	CPT3	DS3	IPCA3
Avg_abs_alpha (%)	1.356	0.830	0.625	1.734
Avg_abs_t_alpha	2.710	1.908	1.330	0.770
GRS p-value	0.00	0.00	0.08	0.00
num_min_abs_alpha	0	3	3	3
Num unexplained alpha	8	4	3	1

³ This is because the data we downloaded from CryptoCompare starts in March 2017, and considering that the construction of certain features may require one year of prior data, the sample period thus begins in March 2018.